

Snowflake Architecture



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Snowflake Snowpro Certified

Section Introduction–Snowflake Architecture

- What is Columnar Storage?
- Understanding the Snowflake Architecture.
- What is a Virtual Warehouse ?
- LAB–Creating a Virtual Warehouse.
- Summary.

What is Columnar storage ?

There are two ways to store data in a database

- Columnar
- Row

Columnar storage also known as column-oriented storage is a data storage technique that organizes and stores **data blocks by columns** instead of rows.

Columnar storage is used for **Data Warehousing** and Big Data Analytics where query performance and efficient data compression are essential.

Advantages of Columnar Storage

- Improved query performance and data retrieval.
- Efficient data compression and storage saving.
- Less compute needed for processing same amount of data as row storage.
- Use of column metadata to skip entire sections of data in a table leading to time and compute saving.
- Efficient parallel loading of data.



What is Columnar storage

Row Storage

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE
100	STEVEN	KING	SKING	515.123.4567	2003-06-17
101	NEENA	KOCHHAR	NKOCHHAR	515.123.4568	2005-09-21
102	LEX	DE HAAN	LDEHAAN	515.123.4569	2001-01-13

100 STEVEN KING SKING 515.123.4567 2003-06-17	101 NEENA KOCHHAR NKOCHHAR 515.123.4568 2005-09-21	102 LEX DE HAAN LDEHAAN 515.123.4569 2001-01-13
Block 1	Block 2	Block 3

- In row storage data blocks store values for each row consecutively.
- Three blocks will need to be read on SELECT of phone number.
- Reading more blocks means slower query performance and more I/O.
- Single record could span multiple blocks and if record is small multiple records could be inside a single block. This generally leads to inefficient usage of disk space and more I/O.
- Row oriented storage if used in Data warehouses where a subset of columns are read from tables with a large number of fields can expect the performance to be poor due to the need to read from a large number of blocks.

What is Columnar storage

Columnar Storage

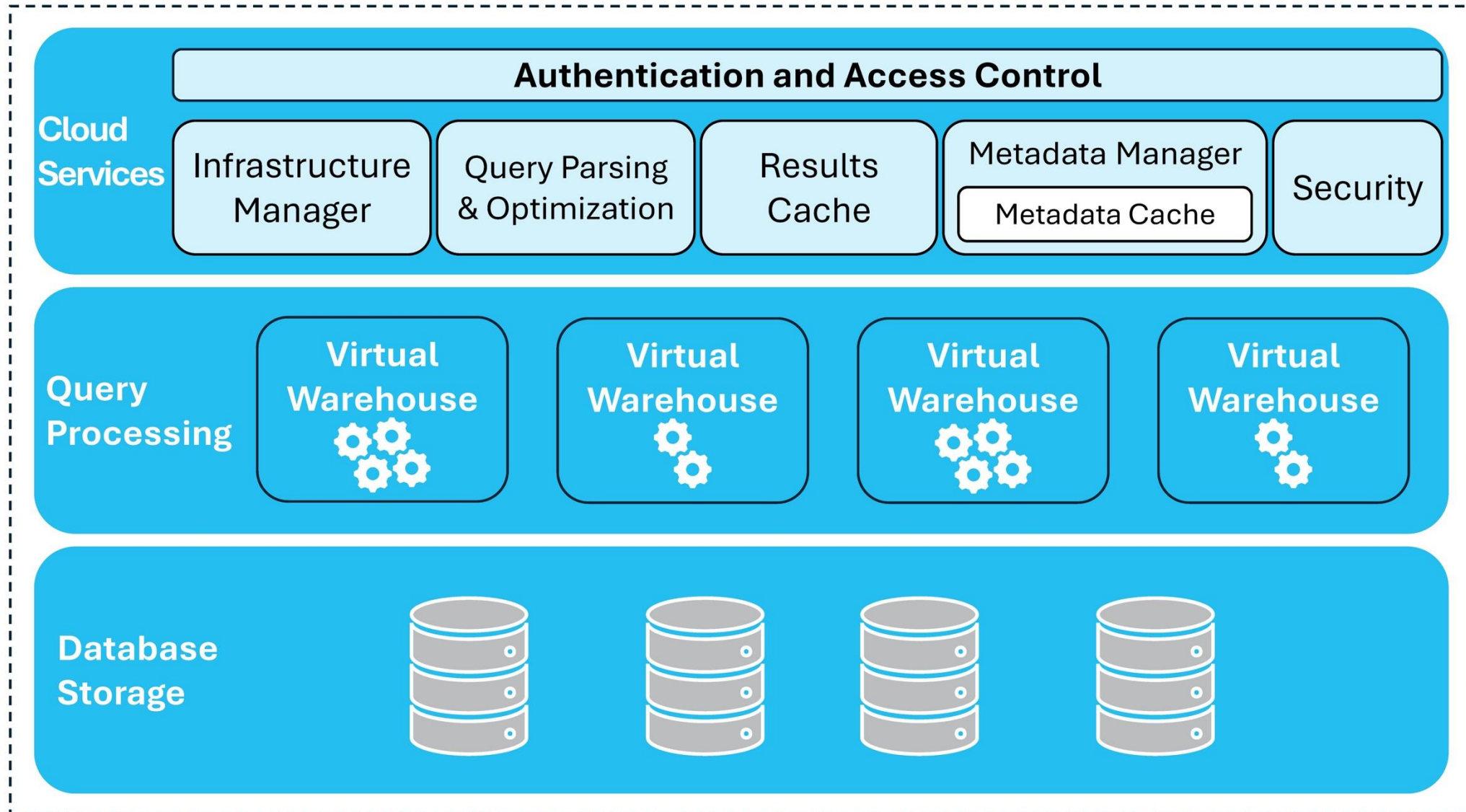
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515.123.4567|515.123.4568|515.123.4569|515.123.4570|515.123.4571|515.123.4572|515.123.4573|515.123.4574|515.123.4575|515.123.4576|515.123.4577|515.123.4578

Block 1

- In columnar storage, each data block stores values of a single column across multiple rows.
- Same operation performed on column storage requires requires $\frac{1}{3}$ the block reads as compared to row storage reducing I/O and time needed to access the data.
- Each block holds the same type of data, data compression of blocks works much better than row storage which has different data types in it further reducing disk space and I/O.
- The savings in space for storing data on disk also carries over to retrieving and then storing that data in memory when data is selected from the table.
- As records are inserted to the table, Snowflake converts the data to columnar storage for each of the inserted rows.

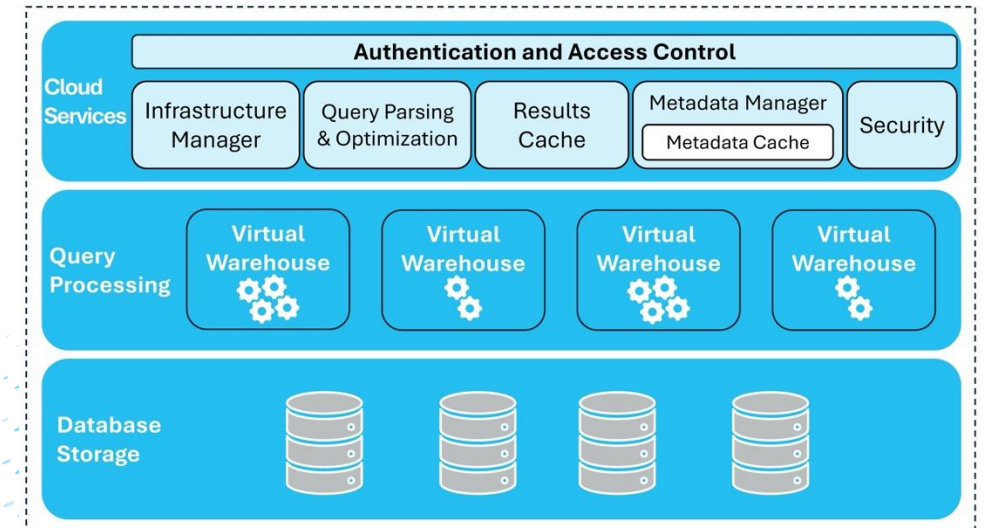
Understanding Snowflake Architecture



Understanding Snowflake Architecture

Database Storage

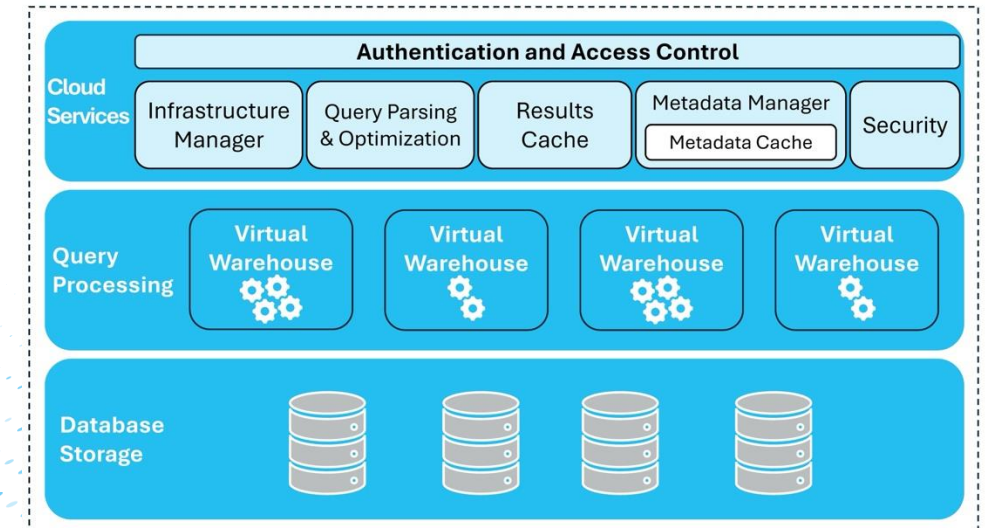
- Data inserted into snowflake tables are reorganized into columnar format and compressed after which it is stored in database storage layer.
- Data stored in the database storage layer is kept on AWS s3 or the cloud provider you choose when you created your account.
- Snowflake manages all aspects of how this data is stored including the distribution compression and metadata storage to name a few.
- Data inside tables in Snowflake are stored in micro-partitions.
- Data inserted into Snowflake tables are automatically divided into micro-partitions each of which contains between 50 MB and 500 MB of uncompressed data.
- Micro-partitions stored by Snowflake are not directly visible or accessible to the data engineer.
- Data stored in the tables in Database storage can be accessed using SQL.
- Max capacity of Snowflake database depends on the storage capacity of AWS s3 which is unlimited hence Snowflake database capacity is unlimited.



Understanding Snowflake Architecture

Query processing

- Query execution or SQL execution is performed in the Query processing layer.
- Snowflake processes queries using “virtual warehouses”.
- Virtual warehouses have nothing to do with data warehouses.
- Virtual warehouses is the processor used to run SQL queries operations like SELECT, INSERT to name a few in Snowflake.
- Snowflake relies on AWS EC2 instances to create virtual warehouses and provide compute for query processing.
- The process of provisioning an EC2 instance when a Virtual Warehouse is created to provide compute is completely managed by Snowflake.



Understanding Snowflake Architecture

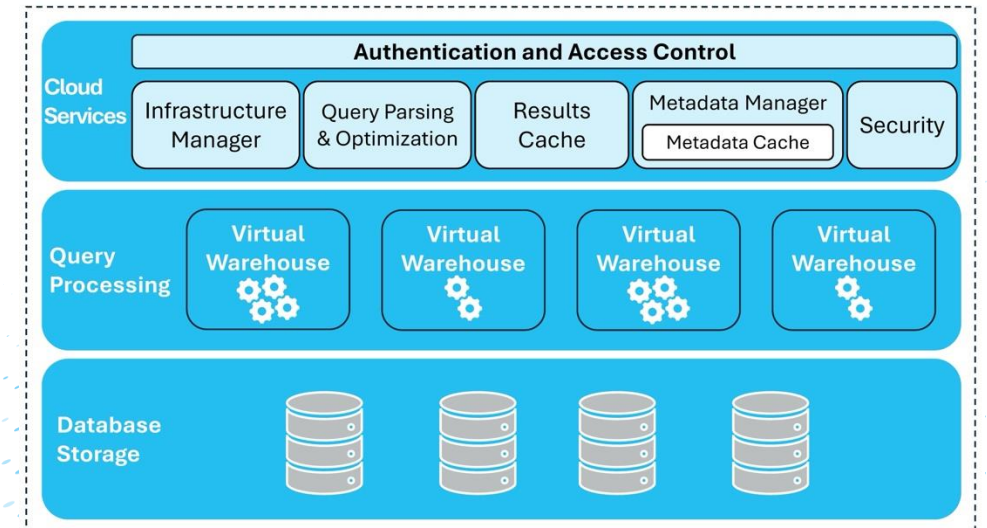
Cloud Services

Infrastructure manager

- Provisioning of virtual warehouses is done by the Infrastructure manager.
- Scaling(size change) of virtual warehouses is managed by the Infrastructure manager.
- Provisioning networking components needed to communicate with cloud service providers.
- Provisioning storage on AWS s3 for table data.
- The Infrastructure Manager works to optimize the use of cloud resources ensuring that they are used efficiently and cost-effectively. This includes optimizing network configurations, storage utilization and compute performance.

Example of Infrastructure manager storage optimization

Deleting table data in s3 that is no needed in Snowflake thus recovering storage space and saving cost.

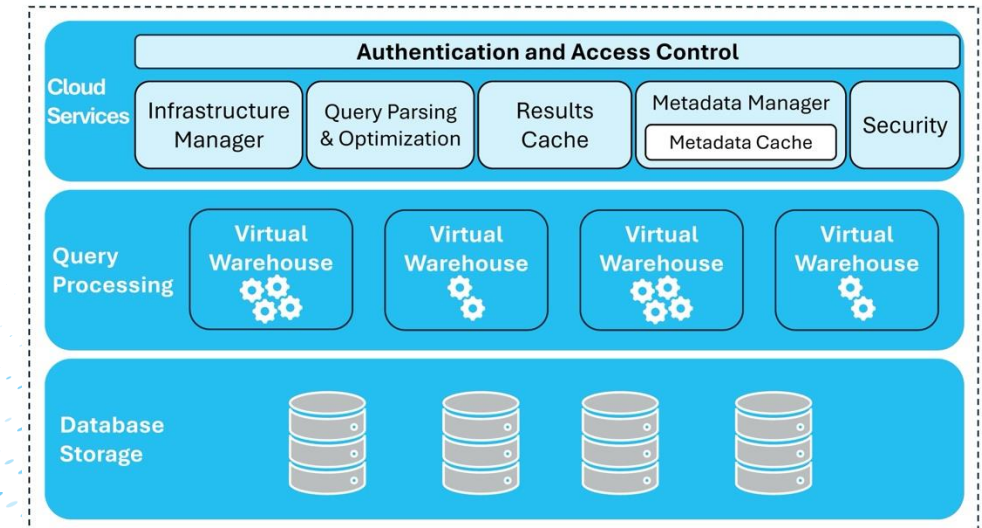


Understanding Snowflake Architecture

Cloud Services

Query parsing and Optimization:

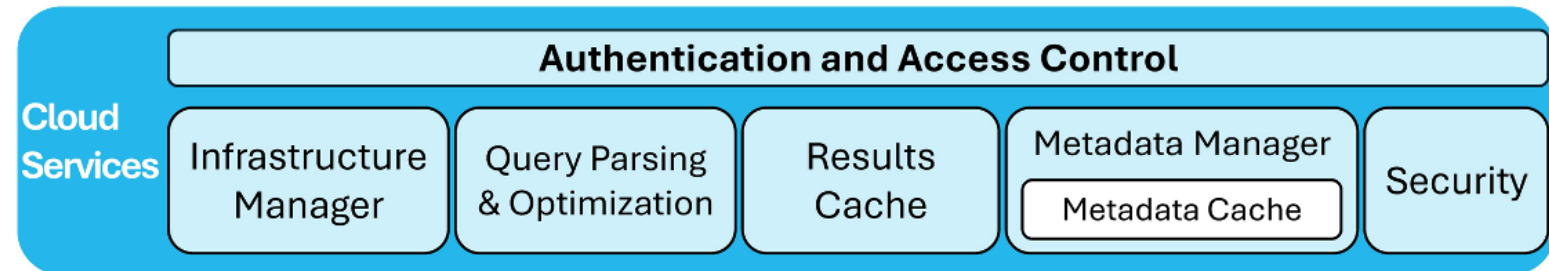
- Syntax checking is done to verify correctness of SQL syntax.
- Execution plan to run the SQL is generated.
- Resource allocation to run the query is provided.
- The execution plan is then passed on to the query processing layer for execution



Understanding Snowflake Architecture

Results Cache

- The results cache stores the results of queries that have been executed previously.
- Result cache is reused if an identical query is submitted again and the underlying data hasn't changed, Snowflake can return the results directly from the result cache bypassing the need to re-execute the query.
- Cached results are stored for 24 hours by default but the cache is invalidated if the underlying data changes.
- Fetching data from the cache saves credits and is much faster.
- For the results cache to be reused the query needs to be identical to a previously run query, even the order of the columns and the case of the query must not change.
- SQL queries that use Non-Deterministic functions like `CURRENT_TIMESTAMP` and `RANDOM` will not re-use the cache.



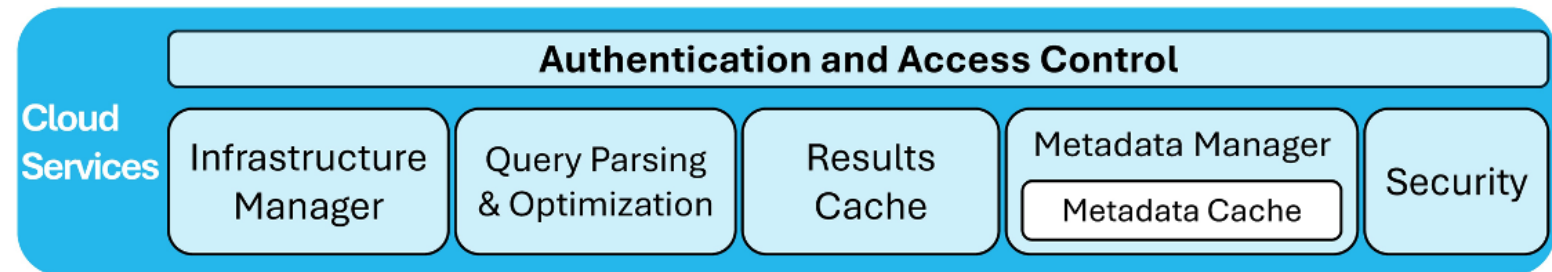
Note:
Non-Deterministic function is a function which provides a different result every time it is run. Since the time is always changing the result of `CURRENT_TIMESTAMP` will also change making it a Non-Deterministic function.

Understanding Snowflake Architecture

Cloud Services

Metadata Manager

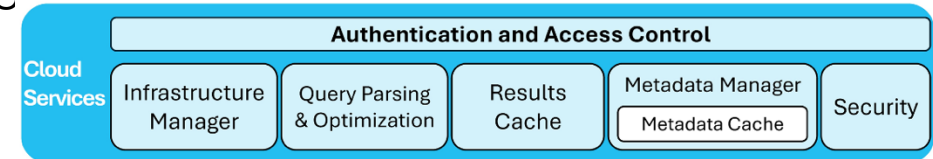
- Stores and manages metadata information like table names and their associated column names inside the metadata cache.
- Table statistics like number of rows in the table.
- Row count like `SELECT COUNT(*)` from `<table>` are made available by the metadata manager from the metadata cache instead of the table.
- Object dependencies are stored & managed.
- Metadata manager and cache help speed up query parsing by providing information on table and column names.



Understanding Snowflake Architecture

Security

- Security in Snowflake cloud service layer is responsible for “Data Encryption at Rest” .
- Snowflake uses AES256 encryption for data at rest.
- Security in cloud service layer is responsible for encryption of data in transit using TLS.
- Security manages encryption keys used to secure data. This includes key generation, rotation, storage and destruction.
- Security maintains detailed logs of user activities, including logins, data access, query execution and modifications. These logs are essential for auditing and forensic investigations in the event of a security incident.
- Monitors the platform for suspicious activities or potential security breaches in real time. Anomalies or unauthorized access attempts can trigger alerts, enabling administrators to take immediate action.



- Security ensures that data sharing between Snowflake accounts is secure. It manages the permissions and access controls that govern which users or accounts can access shared data, ensuring that data is only shared with authorized entities.
- Security enforces network security policies, such as Virtual Private Cloud (VPC) configurations, IP whitelisting and network access controls. These policies help control and secure access to Snowflake environments over the network.

What is a virtual warehouse ?

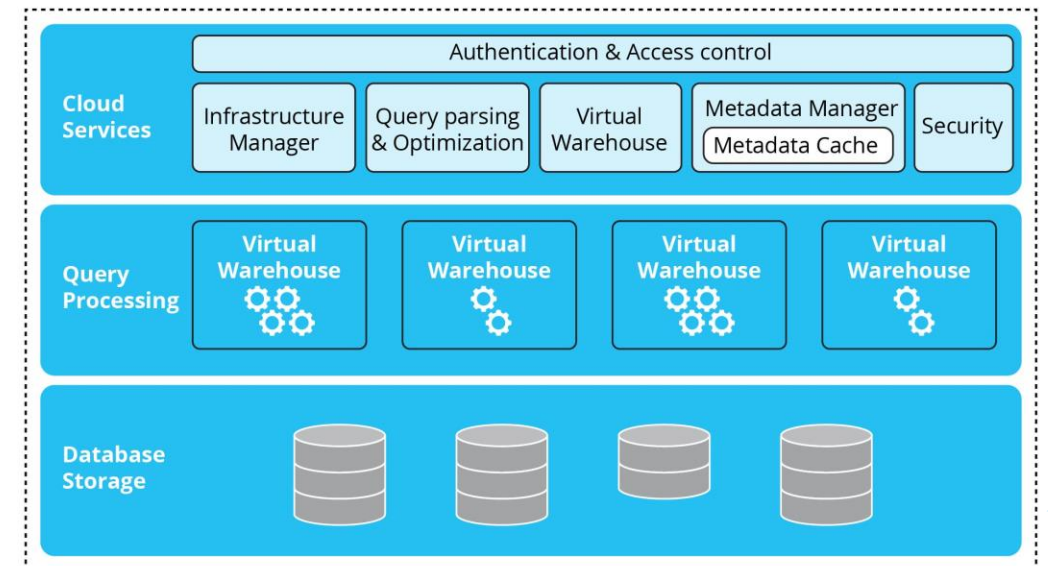
A virtual warehouse which is part of the query processing layer and often referred to simply as warehouse is a compute resource or processor in Snowflake to run SQL queries requested by the user.

Virtual warehouse provides the required resources such as CPU, memory and temporary storage to perform DML and DDL operations like

- SELECT,INSERT,UPDATE,DELETE.
- COPY INTO <>.
- DDL(TRUNCATE, ALTER & DROP).

VIRTUAL WAREHOUSE SIZE

	XS	S	M	L	XL	2XL	3XL	4XL	5XL	6XL
Credits per hour	1	2	4	8	16	32	64	128	256	512



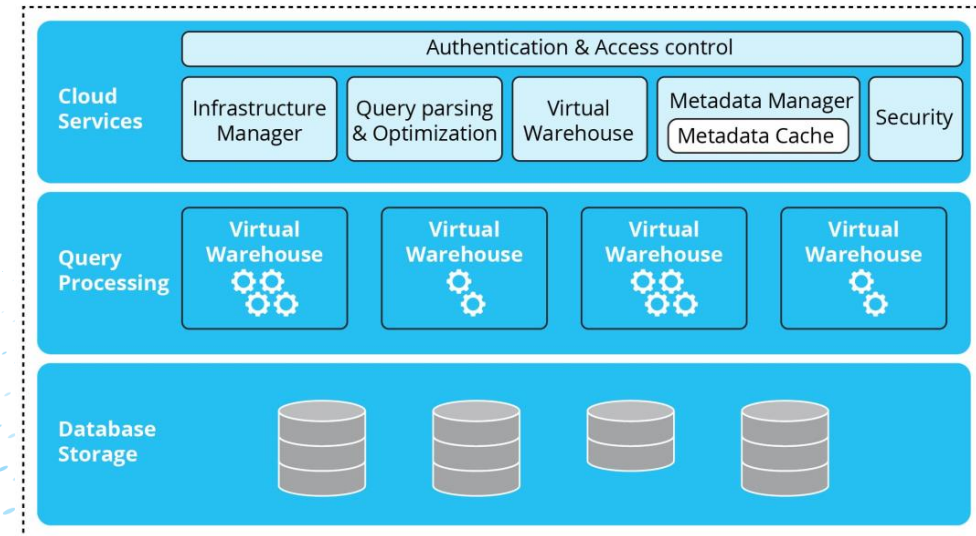
- Virtual Warehouses have properties that can be set to control and automate warehouse activity.

What is a virtual warehouse ?

VIRTUAL WAREHOUSE SIZE

	XS	S	M	L	XL	2XL	3XL	4XL	5XL	6XL
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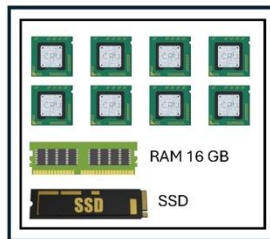
- Snowflake Virtual Warehouses come in 10 sizes. Larger the warehouse more credits it will consume per hour of actively running queries.
- Larger warehouses have more memory and compute capacity and can run queries faster.



What is a virtual warehouse ?

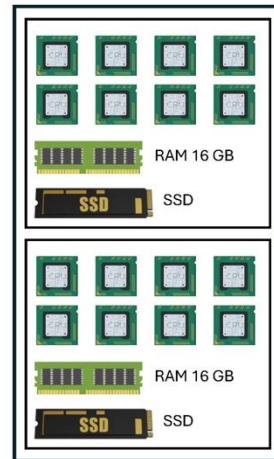
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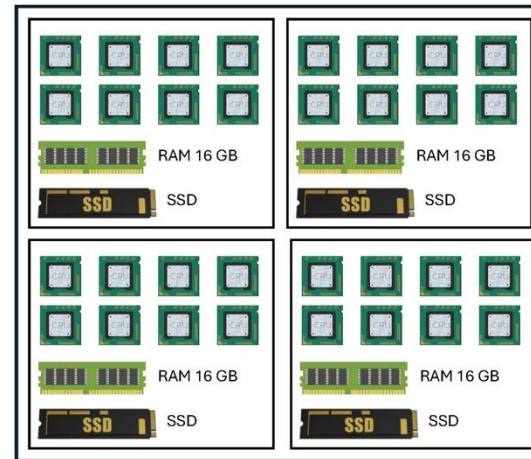
X- Small

- 1 Node
- 8 Cores/Threads
- 16GB RAM
- 200GB SSD



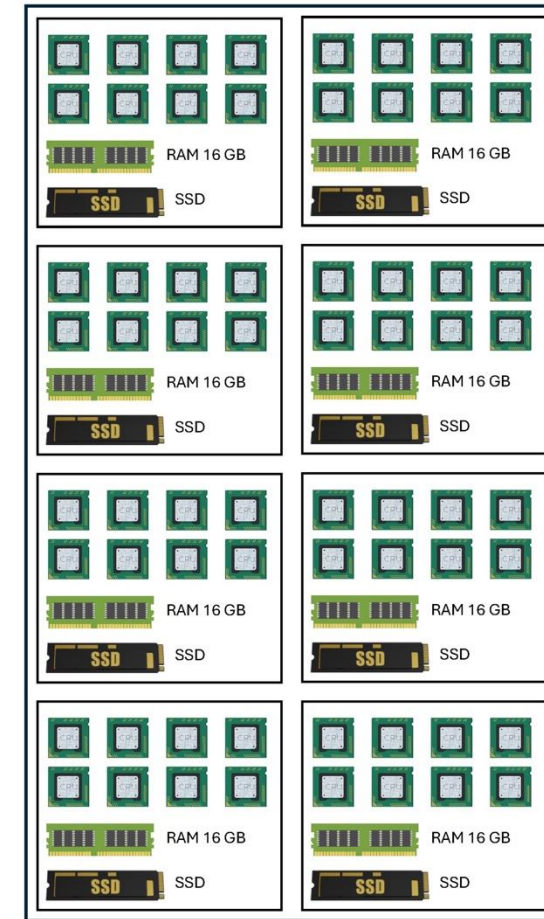
Small

- 2 Nodes
- 16 Cores/Threads
- 32GB RAM
- 400GB SSD



Medium

- 4 Nodes
- 32 Cores/Threads
- 64GB RAM
- 800GB SSD



Large

- 8 Nodes
- 64 Cores/Threads
- 128GB RAM
- 1600GB SSD

Summary

- Snowflake architecture consist of 3 key layers
 - Database storage.
 - Query Processing.
 - Cloud Services.
- Data inserted into Snowflake table are reorganized into columnar format and compressed after which it is stored in the database storage layer which is on AWS s3.
- Data inside tables in Snowflake are stored in micro-partitions which contain 50 to 500 MB of uncompressed data.
- Micro-partitions in Snowflake are not directly visible to the Snowflake data engineer or user.
- Maximum capacity of Snowflake database is unlimited.
- Query or SQL execution is performed in the Query processing layer.
- Snowflake processes queries using virtual warehouses which is compute provided by Snowflake.
- AWS EC2 instances are used to provision virtual warehouses.
- The cloud services layer is a collection of services that coordinate activities across Snowflake database storage and query processing layers. The cloud services is responsible for.
 - Authentication and Access Control.
 - Infrastructure manager.
 - Query parsing & optimization.
 - Results cache.
 - Metadata manager.
 - Security.

Summary

- Infrastructure manager in the cloud services layer is responsible for provisioning and resizing virtual warehouses , provisioning space on s3 for table data and optimizing cloud resources.
- Query parsing and optimization is responsible to verify SQL syntax, generate the execution plan, allocate resources to run the query and pass the query to the execution layer.
- Results cache stores the results of queries that have been executed previously for reuse.
- Result cache is reused if a query is re-run and it meets certain condition which are
 - The underlying data has not changed
 - The order and case of the columns in the query have not changed.
 - It is less than 24 hours since the query was executed
 - There are no non-deterministic functions like CURRENT_TIMESTAMP.
- Metadata manager stores and manages metadata information like table names and associated column names inside the metadata cache.
- Metadata manager and cache improve query performances as the query does not need to go to the database layer to get metadata information.
- The Cloud services layer is sometimes called the “Brain of the snowflake” system as it coordinates all activities in the Snowflake database and account

Summary

- Security in Snowflake cloud service layer is responsible for “Data Encryption at Rest” using AES256 and TLS during transit.
- Security manages encryption keys used to secure data. This includes key generation, Key rotation, storage and destruction.
- Security maintains detailed logs of user activities, including logins, data access, query execution and modifications and these logs are essential for auditing and forensic investigations in the event of a security incident.
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